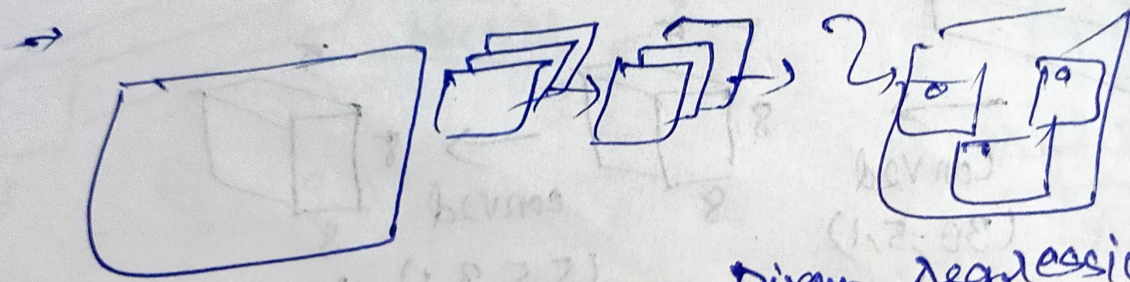
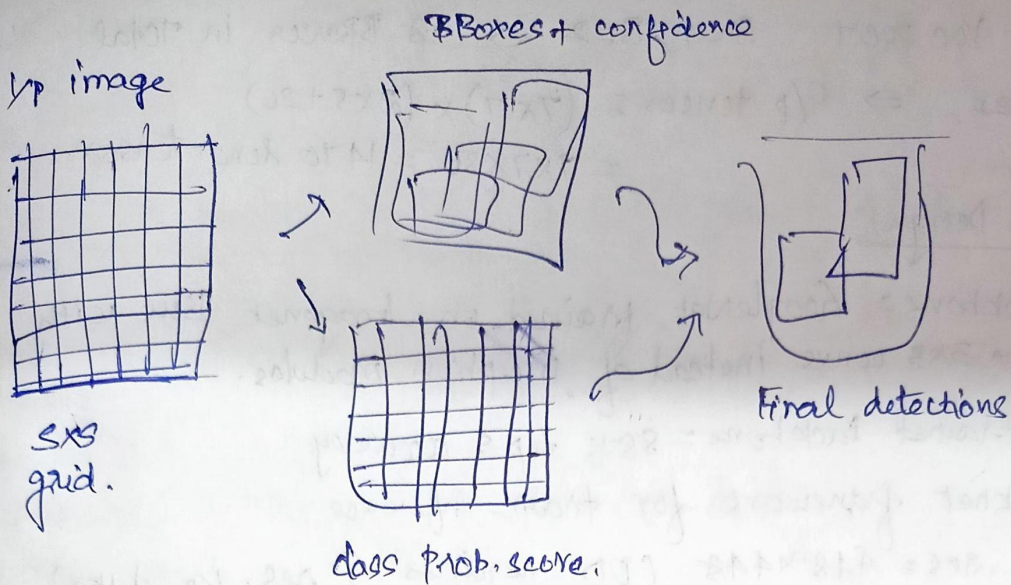


YOLO v11

- Obj. Det as regression problem.
- Focus on RT. $\Rightarrow$ more localisation errors ~~but~~ and ($\downarrow$ mAP)



- It looks globally at ^{entire} image when making predictions
- Less Background errors.
- Difficult to localise small objects.



→ If centre of BBox of obj. falls in grid cell, grid cell is responsible for detecting object.

(normalised with image size)

→ Each grid $\Rightarrow$ 'B' Boxes. Each Box = $[x, y, w, h, \text{confidence}]$

→ confidence = $P(\text{object}) \times \text{IOU truth pred}$

↓
probability of BBox containing object.

↓ Belief in local. ~~error~~ of BBox

$\therefore$ Totally, $S \times S \times (B \times S + c)$

↓ softmax over classes for each grid.

→ w, h are also relative to grid cell (Bounding boxes)

→ $P(\text{Class } i | \text{object}) \times P(\text{object}) \times \text{IOU truth pred}$

= $P(\text{Class } i) \times \text{IOU truth pred}$

conditional prob on object being present

If no. object present, confidence = 0
if object present $\Rightarrow$ IOU score.

PASCAL VOC 2007 $S=7, B=2 \Rightarrow 28$ BBoxes in total
 $C=20, \Rightarrow$ o/p tensor $= (7 \times 7) \times (2 \times 5 + 20)$
 $= 7 \times 7 \times 36 = 1470$ long tensor

Network Design:

- $\rightarrow$ Backbone = GoogLeNet trained on Imagenet But with 3×3 convs instead of inception modules.
- $\rightarrow$ Pretrained backbone = 88% top 5 accuracy
- $\rightarrow$ Darknet framework for train, inference.
- $\rightarrow$ Img. res = 448×448 (DD requires $\uparrow$ res. i/p data)

x, y = offsets of grid location $[0, 1]$

w, h = normalised BBox dims $[0, 1]$

Loss fn: MSE (L2)

- $\rightarrow$ easy to optimize but not ideal
- $\rightarrow$ localisation errors treated same as classification errors.
- $\rightarrow$ Some grids = 0 objects (pushes confidence $\rightarrow 0$) imbalance.

$\therefore$ Soln: Diff. weights for BBoxes containing objects
($\lambda_{coord} = 5$)

& BBoxes not containing objects ($\lambda_{noobj} = 0.5$)

$\rightarrow$ Small error in large BBox = treated same as Error in smaller BBoxes

Soln: $\sqrt{w, h}$ predicted instead of w, h directly.

$\rightarrow$ From each cell, there are 'B' Boxes - so BBox with highest 'IOU' is alone is made responsible for predicting object.

$\rightarrow$ Specialised predictors @ each class, each aspect ratio etc.

Loss fn:

$$x_{coord} \leq \sum_{i=0}^{S^2} \sum_{j=0}^B \sum_{obj} I_{ij}^{obj} [(x_i - \hat{x}_i)^2 + (y_i - \hat{y}_i)^2]$$

$$+ x_{coord} \leq \sum_{i=0}^{S^2} \sum_{j=0}^B \sum_{obj} I_{ij}^{obj} [(\sqrt{w_i} - \sqrt{\hat{w}_i})^2 + (\sqrt{h_i} - \sqrt{\hat{h}_i})^2]$$

$$+ \sum_{i=0}^{S^2} \sum_{j=0}^B \sum_{obj} I_{ij}^{obj} (c_i - \hat{c}_i)^2$$

class
confidence
score

loss only
if object is
present in
that grid and
only for predictor
with highest
IOU

$$+ x_{coord} \leq \sum_{i=0}^{S^2} \sum_{j=0}^B \sum_{obj} I_{ij}^{obj} (c_i - \hat{c}_j)^2$$

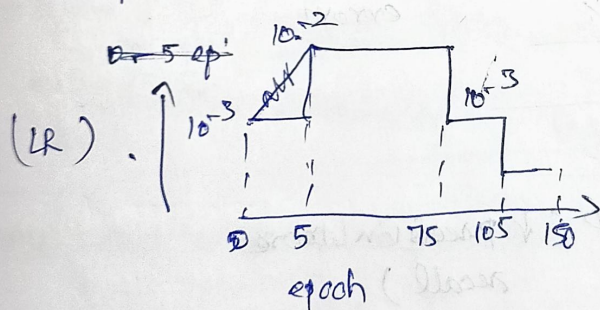
$$+ \sum_{i=0}^{S^2} \sum_{j=0}^B \sum_{obj} I_{ij}^{obj} \leq (\hat{p}(c) - \hat{p}_i(c))^2$$

no object case

class error
only if object
is present

Train Schema:

- 135 epoch (PASCAL VOC 2007 dataset)
- BS = 64, SGD (momentum = 0.9, decay = 0.0005)
- step LR scheduler



leaky ReLU activation,

- Dropout ($p=0.5$)
- Data augmentation

Inference:

- End to end detection
- Grid structure ⇒ spatial diversity in BBox predictions
- NMS post processing

Limitations: Limited BBoxes (2 per grid)

- only 1 class of object per grid
- Data based learning (unlike FCNN variants where selective search ~~is~~ isn't data based)

→ Sparser features compared to RCNN variants.

Error analysis:

→ For each category, top 'N' preds are classified into 1 of 5 categories

→ correct (correct class, $IOU \geq 0.5$)

→ localisation (correct class, $0.1 < IOU < 0.5$)

→ similar (class is similar, $IOU > 0.1$)

→ other (wrong class, $IOU > 0.1$)

→ Background ($0.1 > IOU$ for any object)

All RCNN variants use smaller featuremap
Hence smaller context
↓ Global information
when detecting
Based on proposal

	Fast RCNN	YOLO
Correct	71.6%	65.5%
Localisation	8.6%	19.9%
Similar	4.3%	6.75%
Other	14%	4.0%
BG	13.6%	4.75%

⇒ More localisation errors

→ more FP (↓ precision more recall)